

EVOLUTION OF SATELLITE COMMUNICATION

Over the last 45 years, since the launch of the first Intelsat, communication satellites have advanced phenomenally, changing the way the world communicates. Services have moved up from speech-only to multimedia and there has been a remarkable growth in asymmetrical interactive services and small low-cost terminals

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The potential of satellites for communication was first realised by Arthur C. Clarke, who published a paper in 1945 proposing a three-satellite system to provide worldwide relay communication from the geostationary orbit. The geostationary orbit is unique, as its orbit period of 24 hours precisely matches the earth's rotation allowing a single satellite to provide fixed coverage over a large region. The first satellite to attain geostationary earth orbit (GEO) after two failures was SYNCOM III launched on August 19, 1964. The first Indian geostationary satellite was an experimental communications satellite, Apple, launched on June 19, 1981.

The breakthrough provided by satellites in telecom resulted in a major R&D effort in all related technologies. During the last four decades, satellite communication evolved phenomenally and its applications cover every aspect of human life today. During the early days the emphasis was on international or long-distance point-to-point telecom applications. Later, the application of satellite communications extended to direct satellite broadcast, mobile communication, personal communication and digital broadband communication. Satellite communication is unique, as it is the only means of covering a large geographical area at a reasonable cost,



Sputnik I—the first artificial earth satellite launched by former Soviet Union

and the link can be set up quickly.

Early experiments

Man-made satellite era began in October 1957 when the former Soviet Union launched the first artificial earth satellite, Sputnik I, in a low-earth orbit. It carried a beacon and did not have any communication capability. In response to this, the US launched Explorer I in January 1958. However, Project SCORE (Signal Communications Orbit Relay Equipment) was the world's first communication satellite launched by the US in December 1958; it relayed from

space the voice of the US President Eisenhower giving his Christmas message. All these early satellites were in low-earth orbits with an orbit period of around 100 minutes. SCORE satellite provided store-and-forward kind of communication. The message was sent up to the satellite, recorded onboard and sent down when passing over the desired area of coverage.

In the early experiments, large aluminium-coated inflated balloons were used as passive reflectors for communication signals; they were Echo I (1960) and Echo II (1964). These satel-